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0178/04

October/November 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Electronic Calculator
Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown below that question.

Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π use either your calculator value or 3.142.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 130.

This document consists of **21** printed pages and **3** blank pages.



1 A straight line K joins points $A(-1, a)$, $B(1, 1)$ and $C(3, -5)$.

(a) Given that B is the midpoint of AC , find the value of a .

Answer (a) $a = \dots\dots\dots$ [2]

(b) Calculate the gradient of the line K .

Answer (b) $\dots\dots\dots$ [2]

(c) Another straight line Q which is parallel to line K has equation $y = mx + 1$.
State the value of m .

Answer (c) $m = \dots\dots\dots$ [1]

(d) A point D has coordinates $(d, 9)$.
Given that the length of the line joining B and D is 10 units, find two possible values of d .

Answer (d) $d = \dots\dots\dots$ or $d = \dots\dots\dots$ [3]

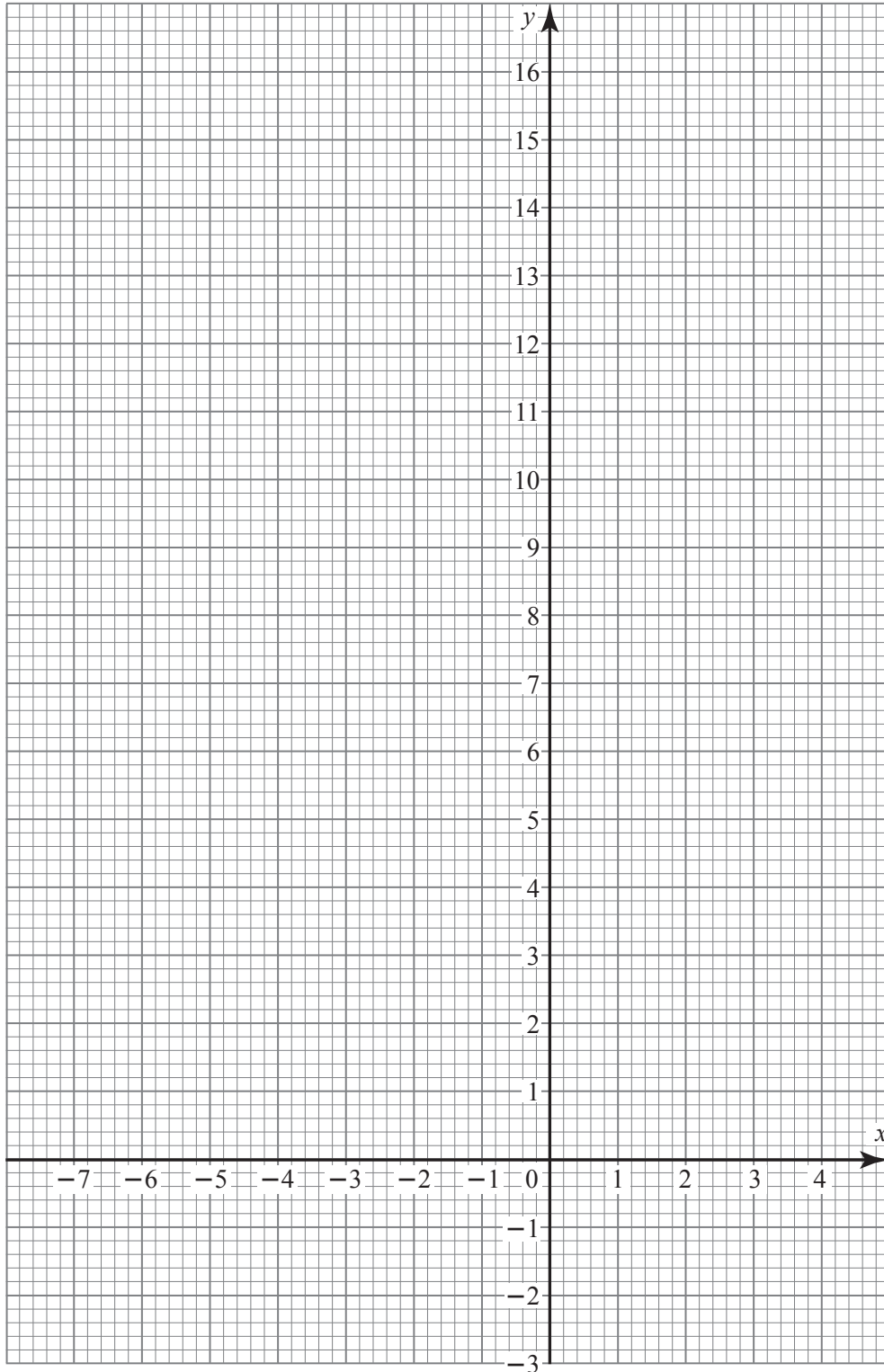
- 2 The table shows some values of x and y for the function $y = 2^{(3+x)}$.

x	-5	-4	-3	-2	-1	0	1
y	0.3		1	2	4	8	16

(a) Complete the table.

[1]

(b) On the grid provided plot the points in the table and join them with a smooth curve.



[3]

- (c) Use your graph to find the value of x when $y = 5$.

Answer (c) $x =$ [1]

- (d) By drawing a tangent, find the gradient of the curve at $x = -1.5$.

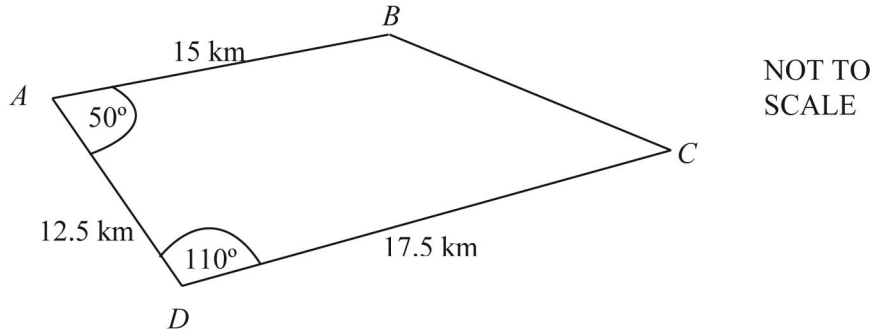
Answer (d) [3]

- (e) On the same axes draw the graph of $y = 3x + 9$. [2]

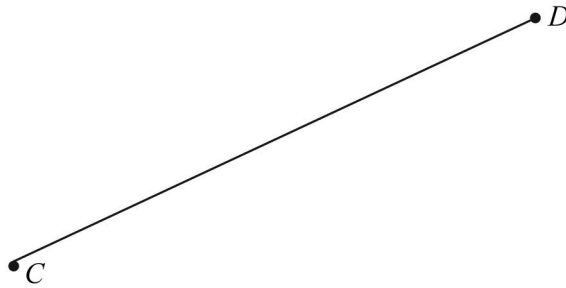
- (f) Use your graphs to find the solution to the equation $2^{(3+x)} - 3x - 3 = 6$.

Answer (f) [2]

- 3 The diagram shows a map of four villages A , B , C and D .



- (a) Using a scale of 1 cm to represent 2.5 km, make an accurate drawing of the map on the space provided. CD has been drawn for you.



Scale 1 cm to 2.5 km

[4]

- (b) Measure and write down the size of the angle ABC .

Answer (b) $^\circ$ [1]

- (c) Find the **actual** distance from village B to C .

Answer (c) km [2]

- (d) Describe the locus of all points that are 2 cm from the line CD .

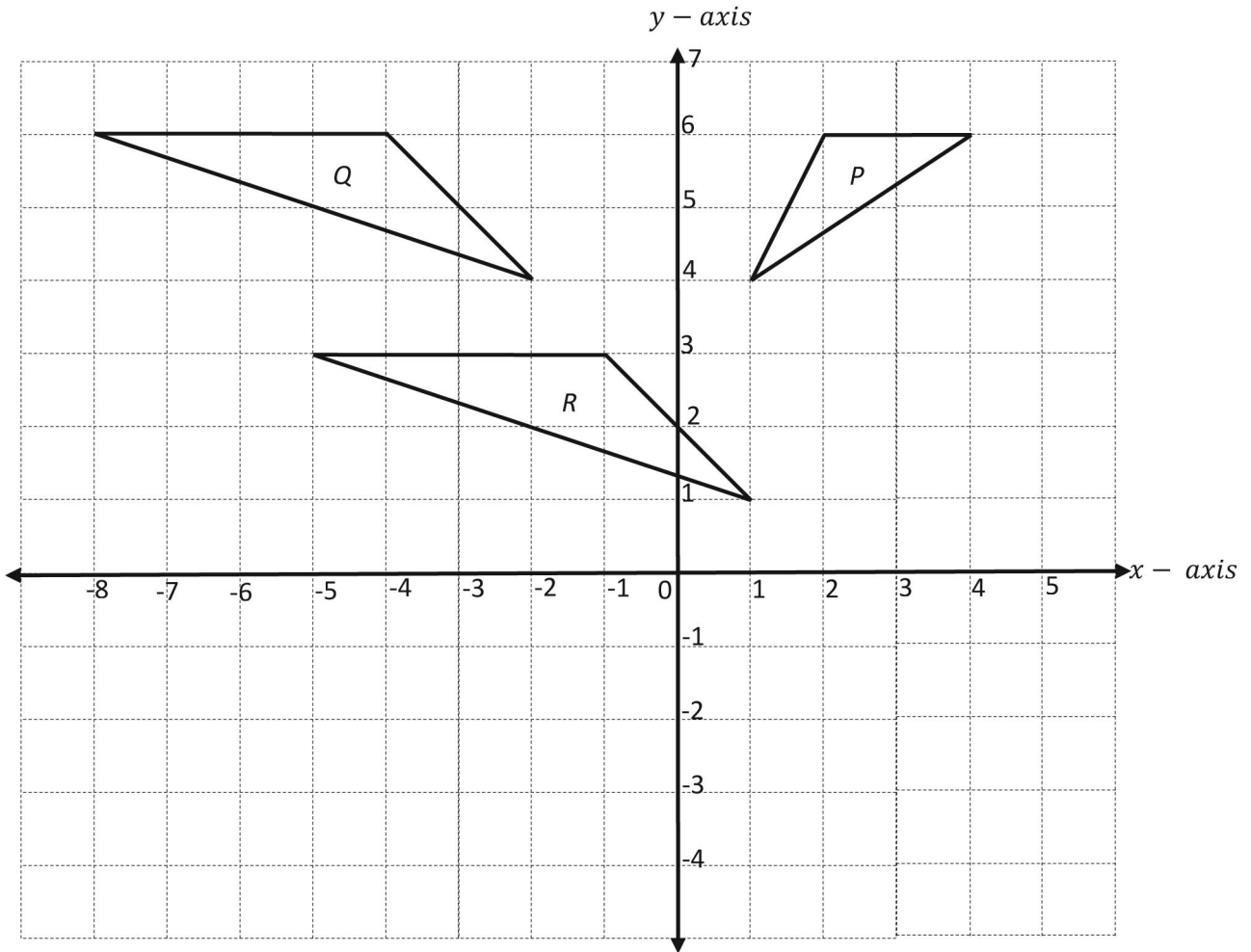
Answer (d)
..... [2]

- (e) Another village E is located on the map such that,

- it is 7.5 km away from village B
- equidistant from BC and CD
- closer to A than C .

On your scale drawing, find by construction, the location of the village E . [3]

- 4 Triangles P , Q and R are shown on the diagram.



- (a) (i) Describe fully the **single** transformation H , that maps triangle P onto Q .

Answer (a)(i) [3]

- (ii) Describe fully the **single** transformation K , such that $KH(P) = R$. SS A* AOB

Answer (a)(ii) [2]

- (iii) $(hx + k, y - k)$ is the image of (x, y) under the transformation KH .

Find the values of h and k .

Answer (c) $h = \dots\dots\dots k = \dots\dots\dots$ [3]

- (b) The matrix of transformation $\begin{pmatrix} -1 & 0 \\ 2 & -1 \end{pmatrix}$ maps triangle P onto U .

Draw and label triangle U .

[2]

- 5 Mpho records the growth in the height of a tree over a period of 23 weeks. At the start, the tree has a height of 0.78 metres tall.

The table shows the height of the tree at the end of each of the seven weeks.

End of week number	1	2	3	4	5	6	7
Height of tree (m)	0.8	0.83	0.87	0.91	0.93	0.95	1.0

- (a) By how much has the tree grown during week 5?
Give your answer in centimetres.

Answer (a)..... cm [2]

- (b) In which week is the maximum growth recorded?

Answer (b)..... [1]

- (c) From Week 8 to Week 23 Mpho summarises the amount of growth in the height of the tree in the table below.

Amount of growth each week (r cm)	$1 < r \leq 2$	$2 < r \leq 2.5$	$2.5 < r \leq 3$	$3 < r \leq 3.5$	$3.5 < r \leq 4.5$	$4.5 < r \leq 5$
Number of weeks	2	1	3	3	5	2

- (i) State the modal class.

Answer (c)(i) [1]

- (ii) Calculate an estimate of the mean growth each week.

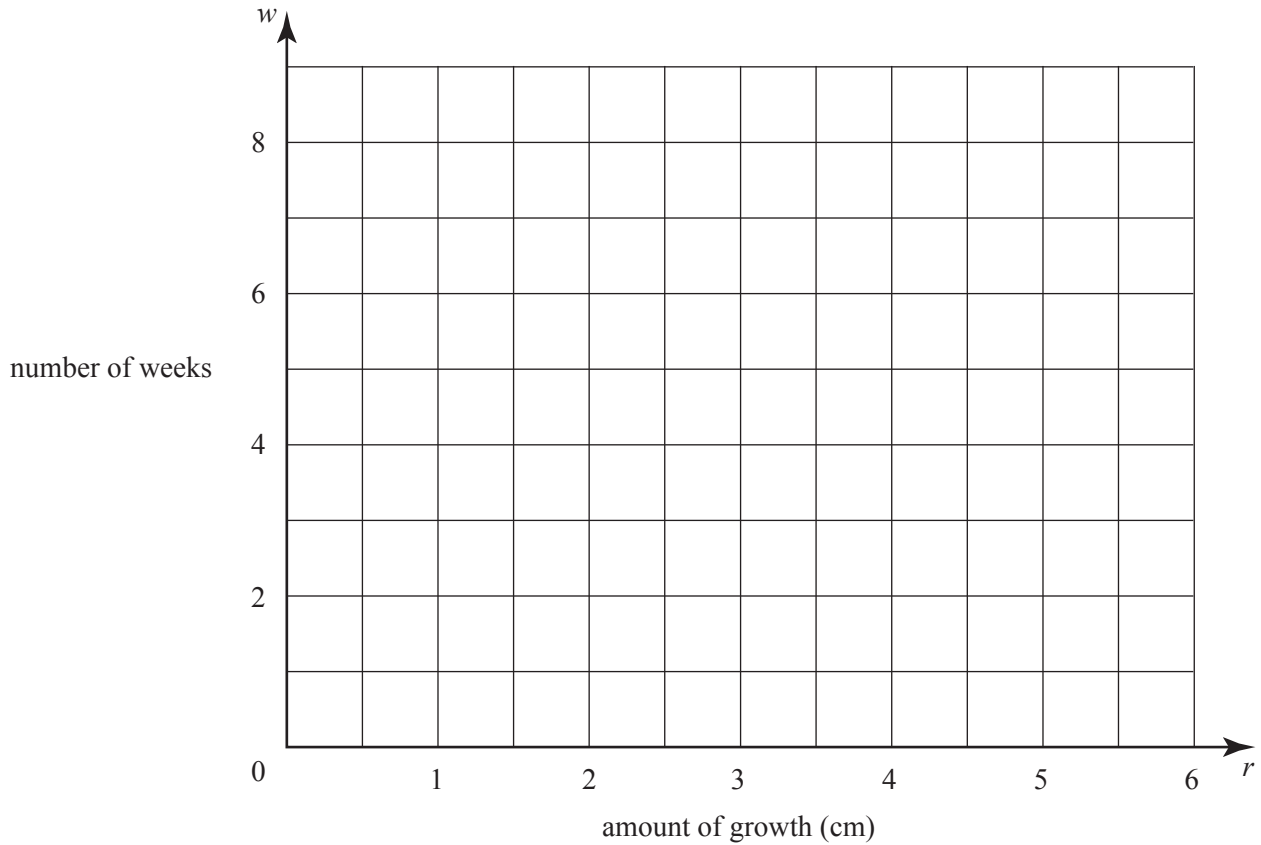
Answer (c)(ii) [4]

(d) (i) Using the tables in part (a) and (c), complete the table below to cover the entire period of 23 weeks.

Amount of growth each week (r cm)	$1 < r \leq 2$	$2 < r \leq 2.5$	$2.5 < r \leq 3$	$3 < r \leq 3.5$	$3.5 < r \leq 4.5$	$4.5 < r \leq 5$
Number of weeks	5	1	4	3		

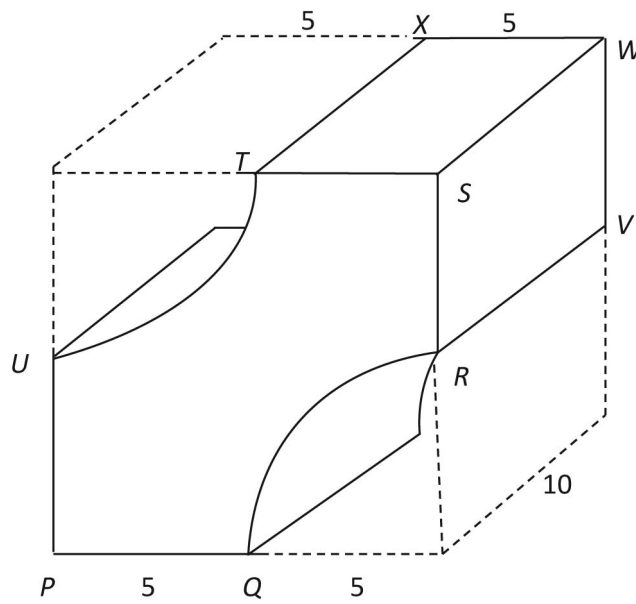
[2]

(ii) On the grid provided, draw the frequency polygon to represent the information.



[3]

- 6 The diagram shows a prism formed by removing 2 quadrant shaped prisms of radius 5 cm from a cube of length 10 cm.



- (a) Find, in terms of π , the area of the cross-section (at the end) of the prism.

Answer (a) cm^2 [2]

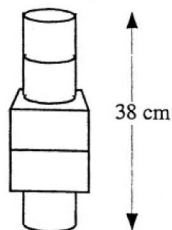
- (b) Calculate the volume of the prism.

Answer (b) cm^3 [1]

- (c) Calculate the total surface area of the prism.

Answer (c) cm^2 [3]

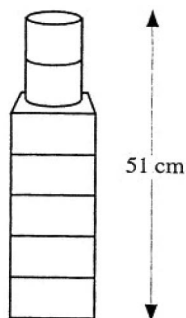
- 7 Neo builds toy towers using identical cylinders and identical cuboids.
A tower of height 38 cm is built from three cylinders and two cuboids as shown below.
Let x cm be the height of one cylinder and y cm be the height of one cuboid.



- (a) Form an equation connecting x and y .

Answer (a)(i) [2]

- (b) Neo then builds the second tower of height 51 cm using **two** cylinders and **five** cuboids as shown.



This new information is represented by an equation

$$2x + 5y = 51$$

Use the above equation and the one in (a)(i) to find the height of a cylinder and a cuboid.

Answer (a)(ii) Height of cylinder = cm

Height of cuboid = cm [4]

- 8 (a) Pule has 3 identical cuboids, each has length l cm and width w cm.

He places the 3 cuboids together on a flat surface to make the shape shown below.

The distance around the base of this shape is given by an equation $6l + 2w = 72$, while the area of its base is given by an equation $3lw = 243$.



- (i) Using equations for the distance and the area, form an equation in w and show that it reduces to $w^2 - 36w + 243 = 0$.

[3]

- (ii) Solve the equation in (b)(i).

Answer (b)(ii) $w = \dots\dots\dots$ or $\dots\dots\dots$ [2]

- (iii) Find the length of the cuboid.

Answer (b)(iii) $\dots\dots\dots$ cm [2]

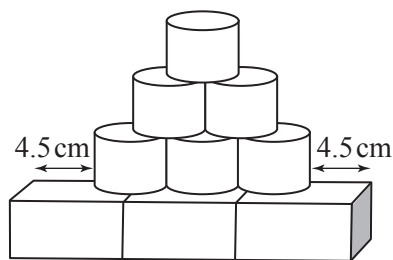
- (iv) The height of the cuboid is 4 cm.

Calculate the volume of each cuboid.

Answer (b)(iv)cm² [2]

- (d) Pule places 6 cylinders on the three cuboids as shown.

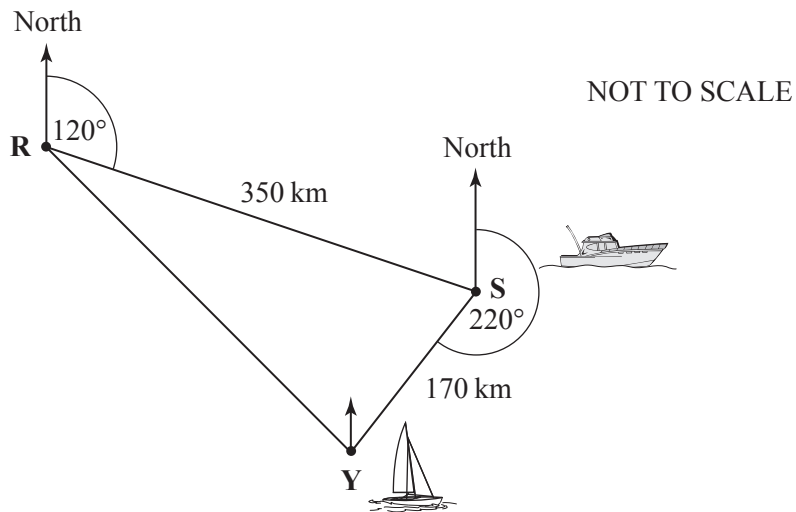
The first row of cylinders leaves an equal distance of 4.5 cm on either side.



Find the diameter of each cylinder.

Answer (c) cm [2]

- 9 The diagram shows a rescue boat **R**, a yacht **Y** and a ship **S**.
S is on the bearing of 120° from **R** at a distance of 350 km.
Y is on the bearing of 220° from **S** at the distance of 170 km.



- (a) Show that angle $RSY = 80^\circ$.

[1]

- (b) Both the yacht and the ship need help from the rescue boat.

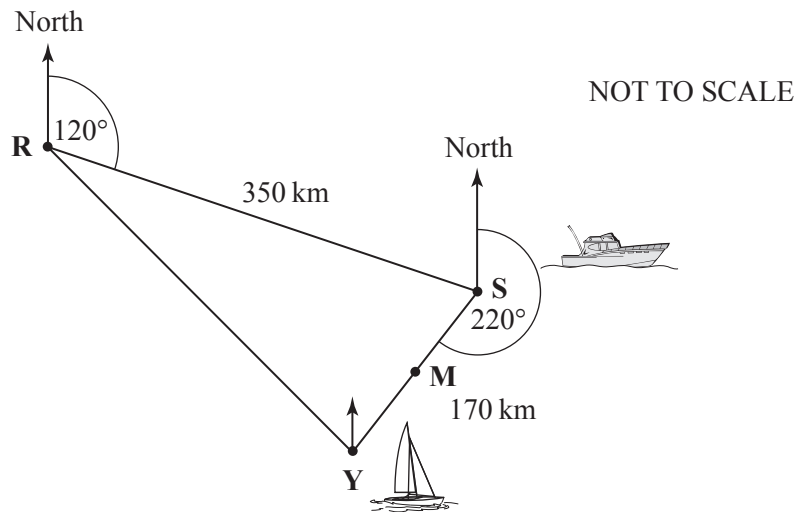
Which one would the rescue boat reach first, the yacht or the ship?
 Show your working clearly.

Answer (d)

because [5]

- (c) The rescue boat radios to the yacht and the boat to sail to a point M, the midpoint of SY.

Point M is 345.5 km away from R as shown.



Find the bearing on which the rescue boat should sail to get to M.

Answer (c) [4]

- 10 (a) The sequence 2, 7, 15, 26, 40, 57, ... has n th term $\frac{1}{2}n(3n + 1)$.

- (i) Find the 20th term of the sequence.

Answer (a)(i) [1]

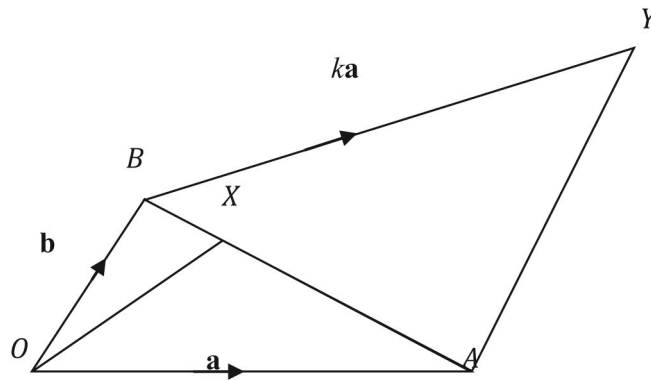
- (ii) Find the position of the term 3775 in the sequence.

Answer (a)(ii) [3]

- (b) Find the n th term of the sequence
7, 15, 26, 40, 57, 77,...

Answer (b) [1]

- 11 In the diagram, $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$, $\vec{BX} = \frac{1}{4}\vec{BA}$ and $\vec{BY} = k\mathbf{a}$.



- (a) Express, as simple as possible, in terms of $\mathbf{a}$, $\mathbf{b}$ and/or k

(i) $\vec{AB}$,

Answer (a)(i) [1]

(ii) $\vec{OY}$,

Answer (a)(ii) [1]

(iii) $\vec{OX}$,

Answer (a)(iii) [2]

- (b) Given that OX is parallel to AY , find the value of k .

Answer (b) $k =$ [3]

- 12 A firm produces door frames and window frames.

For one order the firm has 100 hours available to assemble the frames and 150 hours available to polish them as shown in the table.

	Time in hours to produce one unit		Time available (h)
	Door frames	Window frames	
Assembling	1	4	100
Polishing	3	2	150

Let d be the number of door frames produced and w be the number of window frames produced.

- (a) Write an expression, in terms of d and w , for the total number of hours needed to assemble the door frames and window frames.

Answer (a) [1]

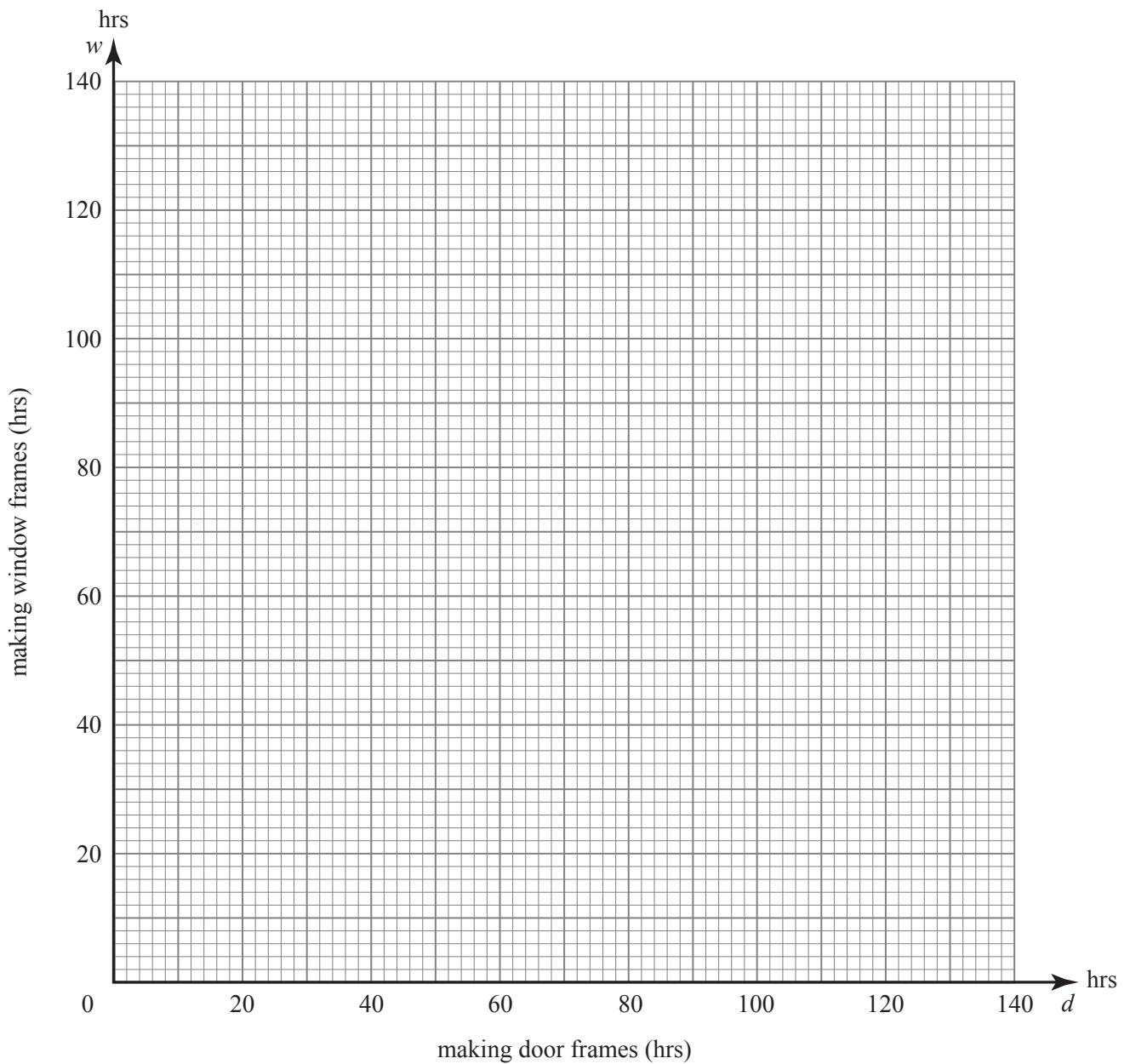
- (b) Two inequalities that represent the information in the table are

$$3d + 2w \leq 150 \text{ and } w \geq 0.$$

Write the other two inequalities.

Answer (b) $x =$
 [2]

- (c) On the grid provided, show the information from part (b) by drawing 4 straight lines and shading the region that is not required.



AL B AOA [4]

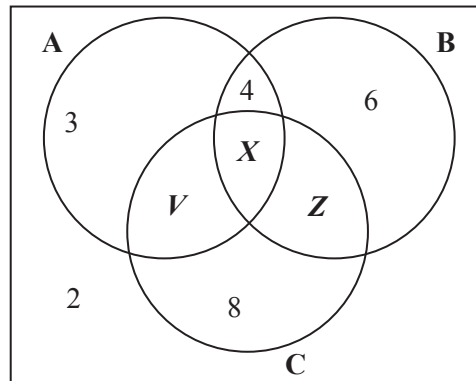
- (d) Given that the firm makes a profit, P in Maloti, where $P = 500d + 700w$, use your graph to find the maximum profit the firm can make.

AL B AOB

Answer (d) M [2]

- 13** The Venn diagram shows the number of students who study Art (A), Biology (B), and Chemistry (C) in one class.

There are 30 students in the class, 14 students study Biology while 10 study Arts.



- (a)** Find the values of x , y and z .

Answer (a) $x =$

$y =$

$z =$ [3]

- (b)** Find $n(B \cap C)$.

Answer (b) [1]

- (c)** How many students study at least two subjects?

Answer (c) [1]

- (d)** Describe the set in $n(A \cap B' \cap C')$ in words.

Answer (d) [1]

- (e)** A student is chosen at random from the class.

Find the probability that a student studies Biology only.

Answer (e) [1]

- (f) Two students are chosen at random from the class.
Find the probability that one of them studies art only and the other studies biology only.

Answer (f) [3]

14 $f(x) = 2x^2 - 1$, $g(x) = 5x + 2$, $h(x) = \frac{x-4}{3}$

- (a) Evaluate $f(2)$.

Answer (a) [1]

- (b) Find the positive value of x for which $f(x) = 17$.

Answer (b) $x =$ [3]

- (c) Find $g^2(x)$.

Answer (c) [2]

- (d) Work out $g^{-1}(x)$.

Answer (d) $g^{-1}(x) =$ [2]

- (e) Given that $f(x) = h(x)$ reduces to $Ax^2 + Bx^{-7} = 0$, find the value of A and B.

Answer (e) $A =$ $B =$ [3]
